

Yosub Shin

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Education

University of Hawai‘i at Mānoa, Honolulu, HI — May 2024 – Present

Ph.D. Student, Computer Science (ICS Department)

Recipient of NSF CISE Graduate Fellowship (CSGrad4US), 2024 — \$159K over 3 years

University of Illinois at Urbana-Champaign, Urbana, IL — Dec 2015

Bachelor of Science in Computer Science, *Highest Honors* — GPA: 3.90 / 4.00

Recipient of Presidential Science Scholarship (Korea Student Aid Foundation) (2008)

Research & Competitions

CTB@ICML 2026 Workshop

m2sv: A Scalable Benchmark for Map-to-Street-View Spatial Reasoning

Yosub Shin, Michael Buriek, Igor Molybog. arXiv:2601.19099

EMM-QA@ICML 2026 Workshop

Data-Efficient Curation for Multimodal Reasoning under Fixed Training Protocols

Yosub Shin, Michael Buriek, Boris Sobolev, Pavel Bushuyeu, Vikas Kumar, Haoyang Xu, Samuel Watson, Igor Molybog. arXiv:2601.10922

NeurIPS 2025 – DCVLR (Data Curation for Vision Language Reasoning) Competition

Winner (1st place). Curated a vision-language reasoning dataset achieving top performance across benchmarks. Presented results in person at the NeurIPS competition workshop. Press coverage

Preprints & Under Review

Beyond Audio and Pose: A General-Purpose Framework for Video Synchronization

Yosub Shin, Igor Molybog. arXiv preprint, 2025. arXiv:2506.15937

Projects

Diffusion Policy on SO-101 — Independent project + write-up — Jul – Aug 2026

- Built a full imitation-learning stack on a \$300 arm (LeRobot, Diffusion Policy): teleop and human-gated DAgger collection tooling, 50+ training runs, 0% → ~43% task success
- Showed hand-guided (kinesthetic) demos cannot teach grasping on high-backlash, position-commanded arms; teleoperation improved grasp rate ~20× by capturing the operator’s compensation for mechanical imperfections
- Designed a matched-block grasp benchmark (280 scored trials) measuring a 27% same-state outcome flip rate—evidence that offline metrics (val loss, action error) and small- N evals cannot rank near-peer policies

YAM-UMI Data-Collection Gripper — Open-source hardware release — Aug 2026

- Designed, fabricated, and open-sourced (CAD + 3D print) a hand-worn UMI-style gripper reproducing the I2RT YAM arm’s stock linear gripper—jaw geometry, camera mount, rack-and-pinion jaw coupling—so wrist-camera observations match between demonstration and deployment

- Replaced the servo encoder used by prior hand-worn variants with fiducial sensing, removing back-drive friction and the load-bearing printed linkage; calibrated a dodecahedral ArUco wrist tracker to 0.46 mm RMS at 94 cm (settled holds); ~\$79 BOM

Dexterous Cable Insertion — Intrinsic AI for Industry Challenge — Mar – May 2026

- Built end-to-end LeRobot pipeline (Gazebo, ROS 2) for contact-rich cable insertion—from scripted oracle demos to learned policies reaching ~50% full-insertion rate in sim
- Scaled sim demonstration generation to 1K+ tier-filtered episodes with DART corrective rollouts; parallel collection/eval and ablations over resolution, cameras, action horizons, and control frequency

Bilateral Teleoperation (SO-101) — Mechanical Engineering course project — Spring 2026

- Assembled and tuned dual SO-101 arms; extended LeRobot with bilateral teleop delivering haptic contact feedback, motion assist, and gravity hold on low-cost Feetech servos
- Implemented 60 Hz per-joint controller (current-gated regimes, velocity lookahead) so operators feel follower stalls during teleoperation

koa-cli — Open-source HPC CLI — 2024 – Present github.com/YosubShin/koa

- Multi-cluster Slurm job submission from local workstation with reproducible repo snapshots, run manifests, and Streamlit dashboard; adopted by UH ICS peers

Experience

Scenery, San Francisco, CA — Feb 2023 – May 2024

Senior Software Engineer

- Designed an AI Editor that auto-cuts hours-long raw footage
- Launched AI transcription + auto-subtitles, boosting free-to-paid conversion

JagalJagal, Seoul, South Korea — Jun 2021 – Jan 2023

Founder, CEO

- Built *TeraSync*, reducing video sync time per episode by 90% (20 hours to 2 hours) via GPU parallelism
- Grew from \$0 to \$70K revenue in under one year

Samsara (NYSE: IOT), San Francisco, CA — Feb 2016 – Dec 2020

Senior Software Engineer II

- Joined as 10th engineer; delivered 8+ Fleet features contributing to \$150M+ ARR
- Owned compliance platform architecture for a 20-person team supporting over 10% ELD market share
- Led routing team; deterministic vehicle routing system using FSMs and immutable audit logs (US Patent 11145208)

Skills

- **Programming:** Python, Go, TypeScript
- **Machine Learning & Robotics:** PyTorch; VLMs; imitation learning (ACT, diffusion); sim (Gazebo, ROS 2); LeRobot, bilateral teleoperation
- **Systems:** GPU training; distributed computing; SLURM; HPC
- **Languages:** English, Korean

Awards

HackIllinois, Urbana, IL — Mar 2015

OculuScooter; 1st place hardware hack among 900 participants